

Roll No.

TOE-801

B. TECH. (EE) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

EXPERT SYSTEM AND FUZZY LOGIC

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) What is fuzzy logic ? Explain discrete and continuous fuzzy set with examples. Also, explain membership value. (CO4, CO3)
- (b) We have a universe comprised of three elements, $X = \{a, b, c\}$, so the cardinal number is 3. What is its power set and cardinality of the power set ? (CO3, CO4)
2. (a) For flight simulator data the determination of certain changes in operating conditions of the aircraft is made on the basis of hard breakpoints in the Mach region. Let us define a fuzzy set on represent the condition of 'near' a Mach number of 0.74. Further, define a second fuzzy set to represent the condition of 'in the region of' a Mach number

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of 0.74. In typical simulation data a Mach number of 0.74 is a hard breakpoint. (CO2)

$$\underline{A} = \text{near Mach } 0.74 = \left\{ \frac{0}{0.730} + \frac{0.8}{0.735} + \frac{1}{0.740} + \frac{0.6}{0.745} + \frac{0}{0.750} \right\}$$

$$\underline{B} = \text{in the region of Mach } 0.74 =$$

$$\left\{ \frac{0}{0.730} + \frac{0.4}{0.735} + \frac{0.8}{0.740} + \frac{1}{0.745} + \frac{0.6}{0.750} \right\}$$

For these two fuzzy sets find the following :

(i) $\underline{A} \cup \underline{B}$

(ii) $\underline{A} \cap \underline{B}$

(iii) $\overline{\underline{A}}$

(iv) $\underline{A} \mid \underline{B}$

(v) $\overline{\underline{A} \cup \underline{B}}$

(vi) $\overline{\underline{A} \cap \underline{B}}$

(b) Draw the block diagram of fuzzy logic controller. Explain the function of each component. Also give a suitable example. (CO2, CO3)

3. (a) A fuzzy set $A(x)$ is given as follows : (CO2, CO3)

$$A(x) = \{(x_1, 0.2) (x_2, 0.3) (x_3, 0.1) (x_4, 0.4) (x_5, 0.4) (x_6, 0.9)\}$$

Find its scalar cardinality and height of $A(x)$.

(b) What is database and rule base in fuzzy logic system. Explain the fuzzy reasoning process with a suitable example. (CO2, CO3)

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4. (a) What is computation. What is soft computing and hard computing.
Mention the characteristic of soft and hard computing. (CO2, CO3)

(b) Explain the following : (CO2, CO3)

(i) Complement of a fuzzy set

(ii) Intersection of fuzzy set

(iii) Union of fuzzy set

(iv) Power of fuzzy set

5. (a) Find the following : (CO2, CO3)

$$A(x) = \{(x_1, 0.1)(x_2, 0.2)(x_3, 0.3)(x_4, 0.4)\}$$

$$B(x) = \{(x_1, 0.5)(x_2, 0.7)(x_3, 0.8)(x_4, 0.9)\}$$

(i) $A(x).B(x)$

(ii) $4.A(x)$

(iii) $A^2(x)$

(iv) $A(x) + B(x)$

(v) $A(x) \oplus B(x)$

(vi) $A(x) - B(x)$

(b) A and B are two fuzzy relations expressed in matrix form. Find the composition of these two fuzzy relations, that is matrix C. (CO2, CO3)

$$A = \begin{bmatrix} 0.3 & 0.2 \\ 0.5 & 0.7 \end{bmatrix}$$

$$B = \begin{bmatrix} 0.3 & 0.2 & 0.7 \\ 0.1 & 0.5 & 0.6 \end{bmatrix}$$

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TOE-805

B. TECH. (EE) (EIGHTH SEMESTER) MID SEMESTER EXAMINATION, March, 2024

ROBOTICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Discuss the Industrial robot anatomy. (CO1)

OR

- (b) Give a classification of robots. Discuss the principle of sensors in brief. (CO1)

2. (a) Discuss the basic objectives of Robotics. Discuss the point of differences between hydraulic and electric drives. (CO1, CO2)

OR

- (b) Discuss the polar coordinate robot configuration. Also, discuss in brief about Pneumatic drives. (CO1, CO2)

3. (a) What is path planning in the context of robotic automation ? (CO2)

OR

- (b) Discuss Range sensors and Proximity sensors. (CO2)

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(2)

4. (a) Discuss the Frenet Frame used for motion projectile. (CO2)

OR

- (b) Explain the working and principle of the stepper motor. (CO2)

5. (a) Discuss the different characteristics of sensors. (CO2, CO1)

OR

- (b) Discuss and list different types of motion control in the context of robotics. (CO2, CO1)

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TEE-803

B. TECH (EE) (EIGHT SEMESTER)
MID SEMESTER EXAMINATION, March, 2024
ADVANCED ELECTRIC DRIVES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Discuss the working of Half-Bridge PWM inverter with suitable diagram. (CO2)

OR

- (b) Explain the working of Single Phase Full-Bridge Inverter. (CO2)

2. (a) Discuss the working of SPWM with Bipolar Switching techniques. (CO3)

OR

- (b) Discuss the working of SPWM with Unipolar Switching technique. (CO3)

3. (a) What do you understand by Generalized Carrier-based PWM inverter ? Explain its functioning. (CO3)

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OR

(b) Explain working of dc/dc buck boost chopper. (CO3)

4. (a) Explain the working of half bridge Voltage Source Inverters. (CO1)

OR

(b) Explain working of Full-Bridge VSI and give its application. What are its limitations ? (CO1)

5. (a) Draw the circuit diagram of Three-phase VSI topology. Enumerate its advantages. (CO4)

OR

(b) Explain static and vector control of an Induction Motor. (CO4)